

Rubin and Rubin Ultra Platforms

#AI

Warren Lau

+852 9181 4766

warren.lau@aletheia-capital.com

Stefan Chang

+852 9250 7088

stefan.chang@aletheia-capital.com

Status Update on Rubin & Rubin Ultra Platforms

This report provides an update on how the development of NVIDIA's Vera Rubin (VR) and VR Ultra platforms is progressing across device and rack-level architecture. For VR, while the Strata module has entered commercial production, NVDA continues to refine both the Vera CPU and Rubin GPU to deliver higher TDP and claimed FP4 compute performance, to strengthen its competitive positioning and reinforce its value proposition. For VR Ultra, both chip-level design and rack-system architecture remain works in progress. The former is expected to tape out in 4Q26, while the latter may be offered in multiple form factors. At the silicon level, both Vera and Rubin are undergoing redesigns and will be paired with a new class of HBM, although bandwidth and density may vary by workload. At the system level, NVDA may support multiple configurations, ranging from dual- and quad-compute racks to as many as eight compute racks, using copper or optical scale-up networking to enable up to 576 GPUs per pod. Besides, NVDA continues to develop the quad-die Rubin GPU SKU and Kyber rack, although the commercialization timeline remains a work in progress.

Vera Rubin Strata in commercial production

Following a brief delay, the VR Strata module entered production in late July. We believe NVDA is still fine-tuning the Vera CPU and Rubin GPU through further tape-outs of both devices, with the objective of reaching a peak TDP of 2,300W and delivering the stated 50PF FP4 inference performance, or roughly 3.3x GB300. By contrast, the standard 1,800W SKU may be capped at 40PF FP4 or less. We expect the enhanced Vera CPU & Rubin GPU to become available in C4Q26.

Vera Rubin Ultra under development

We expect the Rubin Ultra GPU's tape-out in early C4Q26, with the enhanced Vera CPU scheduled for late C4Q26. At the time of writing, we believe NVDA is developing both dual-GPU and quad-GPU Rubin Ultra configurations: the dual-GPU variant appears designed for the existing Oberon NVL72 rack form factor, while the quad-GPU version targets the Kyber system. NVDA is also preparing three HBM variants, differentiated by density, speed, and bandwidth, targeting different AI workloads. Overall, we estimate the VR Ultra's peak TDP at approximately 2,600W. At GTC 2026, NVDA reiterated that the quad-GPU Rubin Ultra with 1T HBM4E can deliver 100PF FP4—twice Rubin's performance—but did not disclose the dual-GPU variant or its FP4 compute capability, even at the recent GTC Taiwan. Given the lower GPU count, we believe the dual-GPU skew will carry materially lower FP4 performance than the 100 FP4 quad-GPU target.

Multi-Oberon racks and Kyber rack systems to coexist

While NVDA continues to develop its quad-GPU Rubin Ultra GPU and the Kyber rack system—engineered to deliver a fourfold compute leap over the current Oberon platform—the sheer complexity of its design and execution may extend commercialization timelines. To bridge this gap and satisfy immediate, insatiable demand for compute, NVDA may introduce multi-Oberon rack architectures, using existing copper or upcoming optical connectivity to scale a unified GPU pool to 144, 288, or up to 576 units in 2H27. In fact, NVDA demonstrated a working prototype of a GB200 multi-rack scale-up system, codenamed Polyphe, at GTC 2026. Based on our industry research, we think this proof-of-concept hardware could be realized in 2H27 paired with the Rubin Ultra GPU, while the complicated Kyber rack will follow suit.

While the scale-up multi-rack systems may offer several advantages—such as a meaningfully higher aggregate bandwidth, greater compute capability, and a lower cost per token—the total sum of investment capex required for this sophisticated hardware may result in a further escalation of capex across neo-cloud providers and hyperscalers from 2027 onward.

Update on Rubin Ultra development

GPU, CPU and packaging

The figures below illustrate the two possible Rubin Ultra GPU designs. The dual-die solution is identical to the upcoming Rubin GPU topology, which consists of two identical GPU tiles and two separate I/O dies. While NVDA will stick with TSMC's N3P process node for Rubin Ultra GPU chiplets, it will redesign the GPU die, likely to enhance agentic AI workloads. Similarly, we expect the I/O tiles to receive a new design, though they will stay at the same process geometries as the current version of the Rubin GPU. The other upgrade is that the reticle size of Rubin Ultra could rise to over 5.5x compared to Rubin's 5x. This is to accommodate the larger size of HBM4E stacks, approximately 10-15% larger than HBM4 used in Rubin 4. We expect the Rubin Ultra tiles to be taped out in 4Q26. Lastly, the Vera CPU is also likely to be redesigned to enhance performance. Overall, we estimate the TDP of the Vera Rubin Ultra (dual-GPUs) to be ~2,600W compared to Vera Rubin's 2,300W (peak) or 1,800W (regular).

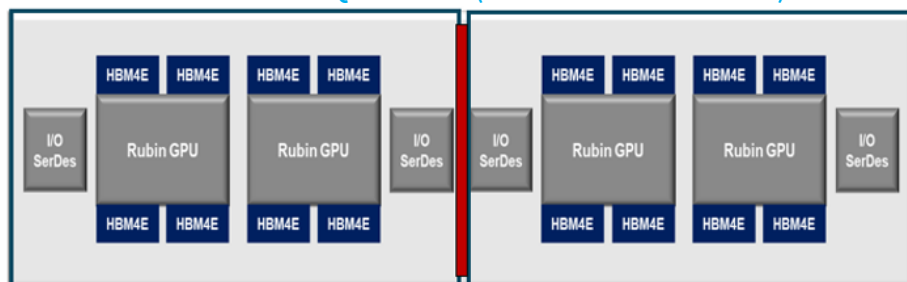
Similar to the Rubin GPU, TSMC will be responsible for the CoWoS packaging of the Rubin Ultra module. Upon the completion of CoWoS packaging, NVDA is likely to employ OSATs to 'bridge' the two Rubin GPU modules via 'stitching' technology as one larger Rubin Ultra GPU or known as the quad-die Rubin Ultra GPU, as shown in Figure 2. The 'stitching' is likely to be performed on the ABF substrate with multiple embedded silicon-bridges that 'connect' the two GPU modules together. We note that because of the relatively large reticle size, it is neither feasible nor economical to package the four-GPU tiles and 16-HBM stacks on a CoWoS form factor. Despite some technical challenges, we note that the quad-die GPU SKU remains in the roadmap (*it hasn't been cancelled*) as it might be used for the Kyber rack form factor.

FIG 1: DESIGN OF RUBIN ULTRA DUAL-DIE GPU



Source: Aletheia/TAG

FIG 2: TOPOLOGY OF RUBIN ULTRA QUAD-DIE GPU (2X RUBIN ULTRA DUAL-DIE GPU)



Source: Aletheia/TAG

HBM density vs bandwidth for training vs inference

At the time of writing, we believe SK Hynix has sent three types of HBM for the qualification of the Rubin Ultra GPU – 192GB of HBM4, 256GB of HBM4E and 384GB of HBM4E. Compared to the Rubin GPU's 288GB of HBM4, the 192GB is one-third less in density and offers no improvement in both speed and bandwidth. We think it is less likely to be chosen, given NVDA is likely to charge a higher price for the Rubin Ultra module (see next section for details). Besides, the design of the reticle size of the Rubin Ultra module remains over 5.5x as of today, suggesting a higher possibility of HBM4E. The second option – 256GB of HBM4E – seems reasonable, with 30%-40% increases in speed and bandwidth offset by an 11% drop in density. The last (384GB of HBM4E) remains an option, although it costs more (see next section), which may prevent NVDA from adopting it.

NVDA's potential reduction of HBM density in the Rubin Ultra could be counter-intuitive to the rapid rise of long context agentic AI workloads as well as the increasingly large size of AI frontier models. For example, a 70bn parameter AI model processing a 1mn token context window requires 140GB of memory density just to load the model and a further 130GB of memory to hold the conversation history.

However, one should note that memory bandwidth is more critical than density for AI training because larger AI models require moving massive amount of data or model weights from a memory device to a GPU to prevent the GPU being idle. As shown in Figure 3, the increase in memory density (the same is also true when density drops) will not impact the memory bandwidth, which is a function of device speed and bus width. For example, 8Hi and 12Hi HBM4E HBM stacks offer identical bandwidth because the speed of both is rated at 14Gbps, while the bus width is at 2,048 bits. So, to further increase the bandwidth, NVDA could demand memory suppliers that they increase the speed to the peak 16Gbps (DRAM process node-related), which could yield as many as 4.1TBps (per stack), over 50% more than HBM4. Alternatively, NVDA can raise the HBM stacks from 8 to 12, increasing the overall system bandwidth. For example, 12 stacks of HBM4E (8Hi or 12Hi) could yield a total 49.2TBps bandwidth, 50% higher than the 8 stacks HBM GPU module.

FIGURE 3: HBM TYPES AND DENSITY FOR THE VERA RUBIN ULTRA PLATFORM

	Rubin Ultra in 2H27			Rubin in 2H26
HBM type	HBM4	HBM4E	HBM4E	HBM4
HBM stacks	8Hi	8Hi	12Hi	12Hi
Die density	3GB	4GB	4GB	3GB
Stack density	24GB	36GB	48GB	36GB
Speed	~10.5Gbps	~14Gbps	~14Gbps	~10.5Gbps
Bandwidth (per stack)	~2.69TBps	~3.58TBps	~3.58TBps	~2.69TBps
Dual-GPU HBM density	192GB	256GB	384GB	288GB
Quad-GPU HBM density	384GB	512GB	768GB	NA

Source: Aletheia/TAG (as of August 2026)

However, the reverse is true for long context agentic inference workloads. In agentic AI, agents operate through multi-step reasoning loops, multi-agent coordination, and tool execution. For example, a single active 128k token context window can demand over 50GB of memory just for the cache of one user session. If a system lacks the memory bandwidth, the agent generates text slightly slower; however, if the system lacks memory density, the agentic pipeline is likely to fail the mission. High bandwidth cannot help if there is no physical space left to store these active tokens.

In recent times, NVDA has proposed several innovative hardware solutions to cater for long context agentic workloads, such as Rubin CPX in September 2025 during the AI Infra Summit. Rubin CPX uses the more cost-effective GDDR7 for the pre-fill (context digestion and KV cache) while leaving the decode work for HBM4. However, this solution seems to be taking a pause (though not cancelled) probably due to the tight memory capacity and the escalating costs.

More recently, NVDA introduced the Inference Context Memory Storage Platform (ICMS) at CES in January 2026. The idea is that instead of forcing precious HBM and SoCAAM to hold massive conversation logs, ICMS pools and treats the KV cache as a shared, rack-scale memory resource. ICMS has since been renamed as Context Memory Storage (CMX) in GTC 2026 and is powered by the BlueField 4 STX storage rack and can house as many as 9,600TB of ultra-fast eSSD memory.

The STX storage rack is linked to the NVL72 compute rack using Spectrum X ethernet switches (CPO could be an option). Using its proprietary software and the connectivity (144x CX9 NICs and 36x BF4 DPU in the NVL72 compute rack and 64x BF4 DPU and 64x CX9 in the STX storage rack), NVDA can offload and onload the KV cache between the compute and storage racks in sub-millisecond latencies (bypassing CPU, SoCAAM and HBM), preventing the GPU from sitting idle.

We believe the leading candidate for the high speed eSSD memory is Kioxia's XL-Flash, which is designed to meet the NVDA CMX and Storage-Next Platform. The company has sent a sample for qualification, and the results are expected to be available in late-3Q/early 4Q. Please refer to our research ([initiation](#), [XL-Flash](#)) for more details.

While the final outcome of the HBM type and density are yet to be determined, we think it could consist of a couple of options. For large-scale frontier AI models, a higher bandwidth HBM4E (but not necessarily a higher density) should be a preferred solution while for long context multiple agentic AI workloads, a combination of a reasonable amount of HBM density (but not necessarily a higher bandwidth) and an ultra-fast speed and high density eSSD could yield a more desirable outcome. This brings the discussion to the VR Ultra rack systems, as can be seen below.

The possible structure of Rubin Ultra racks

We believe NVDA is working on a number of rack systems to better cater for different AI workloads. This includes a unified Kyber rack form factor (with quad-die GPU) which can house as many as 576 GPU tiles as well as multi-Oberon racks (dual-die GPU) which may consist of several compute racks as well as networking and storage racks. Because of the complexity of designs and executions of the Kyber system, the commercialization of this architecture may take longer than expected. As such, using multiple racks on the existing Oberon structure is likely to be the solution that bridges the gap for the Vera Rubin Ultra platform. In addition, the insatiable demand for compute at an economical unit cost of tokens may prompt NVDA to shift from the existing single NVL72 rack to a multi-rack form factor such as dual- or quad- or even as many as eight-racks. Furthermore, the latter offers more flexibility by adding networking and storage racks using NVDA's optical ethernet switch.

In fact, at GTC 2026, NVDA demonstrated a working prototype GB200 multi-rack scale-up system, consisting of eight compute racks – Polyphex. This system was further highlighted at the 2026 IEEE VLSI Symposium forum in June 2026. Based on our discussions with industry experts, we think this proof-of-concept hardware could be realized in 2H27, pairing it with the Rubin Ultra GPU while the complicated Kyber rack could follow suit.

FIGURE 4: PROTOTYPE OF SCALE-UP 576 GPU RACK - POLYPHE



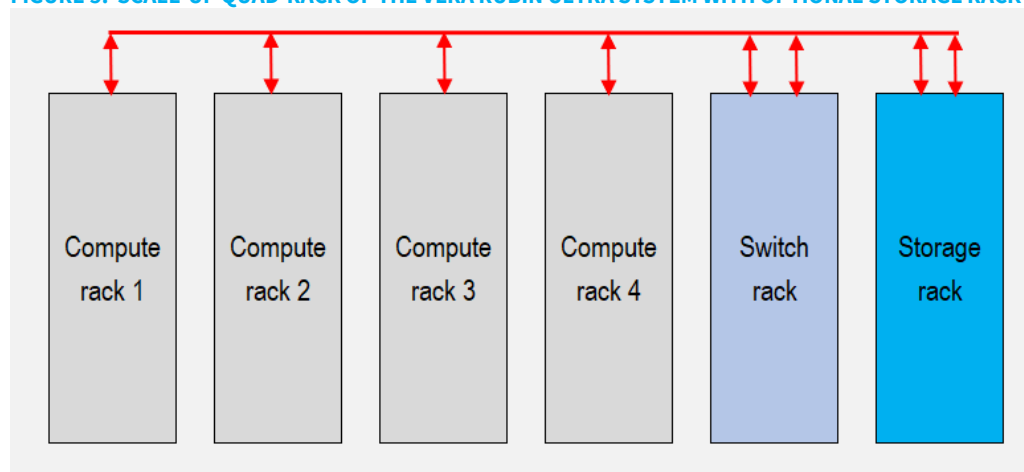
Source: nVidia

One potential benefit of using scale-out for multiple racks is that NVDA can aggregate the total system bandwidth significantly higher than a unified Kyber system. As shown in Figure 6, the quad-rack could command a system bandwidth of over 8PBps, almost double that of the Kyber rack. Even if NVDA adopts the lower grade of 192GB of HBM4, the aggregated system bandwidth would still amount to over 6PBps, 30-40% higher than Kyber's matrix. Since system bandwidth is a crucial factor for larger scale AI model training, *the scale-out*

connectivity for multi-rack could be a viable alternative for a higher number of HBM stacks and more advanced and speedier HBM in one GPU within a single NVL72 compute rack.

The other advantage of the scale-up multi-rack is that NVDA can offer a flexible hardware setup to cater for different AI workloads. In the case of long context multi-agent applications where KV cache is critical, NVDA can offer a dedicated high-speed eSSD storage rack (BF4 STX storage) via the networking rack (Spectrum6 SPX rack) for high-speed KV cache offload and onload with minimum latency. Since the BF4 STX storage can support up to 1.2PB of ultra-fast eSSD (or 16TB density per GPU), which interfaces directly with GPU (bypassing CPU/SoCAAM), this hardware setup could be critical for mission-critical agentic workloads. This is especially true considering the cost of this purpose-built-and-designed ultra-fast eSSD is a fraction of the cost of HBM and server DRAM. In the case of training larger scale frontier AI models, a possible dual-rack system may do the job because the aggregated bandwidth could reach over 4PBps, 2.5x more than a VR200 NVL72 single rack.

FIGURE 5: SCALE-UP QUAD-RACK OF THE VERA RUBIN ULTRA SYSTEM WITH OPTIONAL STORAGE RACK



Source: Aletheia/TAG (August 2026)

FIGURE 6: HBM TYPES AND DENSITY FOR VERA RUBIN ULTRA PLATFORM

VR Ultra platform	Dual-rack	Quad-rack	Eight-rack	Kyber ¹	VR200 NVL72
GPU configuration	Dual-GPU	Dual-GPU	Dual-GPU	Quad-GPU	Dual-GPU
Rubin Ultra GPUs ²	144	288	576	144	72
Total GPU dies/tiles	288	576	1,152	576	144
System HBM density ³	36.8TB	73.7TB	147.5TB	124.4TB	20.7TB
System bandwidth ⁴	4.1PBps	8.2PBps	16.4PBps	4.6PBps	1.58PBps
SoCAAM density ⁵	55.3TB	110.6TB	221.1TB	216TB	27.6TB
Total system density	92TB	184TB	369TB	340TB	48TB
GPU FP4 ⁶	TBA	TBA	TBA	100PF	50PF
System FP4 ⁶	TBA	TBA	TBA	15EF	3.6EF
Scale-up	Copper	Optical	Optical	NA	NA

Note: 1-announced in 2025 GTC and reiterated in 2026 GTC. 2-We use the GPU module as a unit. 3-We assume HBM4E 8 Hi (256GB) as the default HBM. 4-We assume the speed of HBM4E is 14Gbps (bandwidth of 3.58TBps). 5-We assume 768GB of SoCAAM density for each Vera CPU. 6-This refers to inference FP4 compute.

Source: Aletheia/TAG (as of August 2026)

What about the costs?

Figure 7 shows our estimated BOM for the Vera Rubin modules. For the dual-GPU Rubin Ultra Strata module, we forecast this to rise by ~30% in 2H27 over the cost of the Rubin Strata in 2H26. Virtually all the extra costs are driven by device and component inflation, notably HBM and SoCAAM, followed by logic devices and others such as ABF substrate, PMIC and possibly passive components. As a result, memory could account for 70% of the BOM for the Rubin Ultra module, up from the mid-60% for the Rubin Strata module and far more than the mid-40% for the GB300 Bianca module. This may explain why NVDA is actively looking to contain the memory content, first by lowering the density of SoCAAM in the Rubin Strata in 2H26, with a further reduction possibly happening in 1H27. Similarly, the possibility of using a lower density 256GB HBM4E from the previous expectation of 384GB and the originally stated 1T of HBM (first revealed in 2025 GTC) is further evidence.

FIGURE 7: ESTIMATED NVDA's COMPUTE MODULE BILL-OF-MATERIAL AND ASP

Rubin Ultra in 2H27	Dual-GPU	Quad-GPU	Rubin in 2H26	GB300 in 2H25
Logic ¹	17%	18%	22%	36%
HBM (256GB 4E)	21%	20%	19% (288GB 4)	27% (288GB 3E)
SoCAAM (768GB)	51%	49%	45% (768GB)	20% (520GB)
Others ²	11%	13%	14%	17%
Total BOM	\$42,162	\$86,850	\$27,536	\$14,104
Est ASP	\$168,649	\$347,399	\$130,000	\$80,000
Normalized GPU ASP	~\$84,000	~\$87,000	\$65,000	\$40,000

Note -1 includes GPU, CPU & CoWoS. 2-consists of MLCC, ABF substrate, PMIC, mainboard etc

Source: Aletheia/TAG (as of August 2026)

Should NVDA hold on to its 75%-80% gross margin, we estimate the ASP for the Rubin Ultra Strata is likely to top \$170,000 in 2H27, up by 30% over the Rubin Strata in 2H26, which we expect to be priced at around \$130,000. It remains to be seen if the 30% price hike is acceptable to clients given that NVDA has not released the FP4 compute of the dual-GPU (relative to the claimed 50PF of FP4 with 2,300W Rubin). In the case of the GB300, NVDA improved the FP4 by 50%-60% over the GB200, easily validating the ~35% price hike. However, NVDA could justify the rise in prices if the scale-up multi-rack system offers more compute, bandwidth and a lower unit token cost.

This being said, one should note that the overall sum of investment in such a multi-rack system is likely to escalate from 2027/28 onward, potentially forcing neo-cloud operators to seek more external capital. The hyperscalers may also have to fork out for higher capex investment or accelerate their in-house ASIC development to lessen their reliance on merchant GPU suppliers.

Altheia Research Team

Macro/Strategy	LEAD ANALYST	Sector	LEAD ANALYST
Global Strategy	 Jonathan Wilmot	Technology Hardware	 Warren Lau
Asianomics	 Dr. Jim Walker	Consumer & Internet	 Nirgunan Tiruchelvam
Tech Thematic Strategy	 Keith Woolcock	China New Economy	 Eric Wen
Asia Equity Strategy	 David Scott	Asia Telecoms	 Chris Hoare
China Strategy	 Vincent Chan	CrossASEAN	 Angus Mackintosh
Tactical Alpha Strategy	 Justin Collazo	India	 Maulik Patel
Global Commodities	 Steven Schlegel	MENA	 Jaap Meijer

Product Marketing Teams

Serving 200+ clients, 8,000+ touch points, in 20+ geographies with \$15tr+ AUM



● Al Park 1 201 962 0539 al.park@aletheia-capital.com AMERICAS - NEW YORK	● Linda Gustafsson 44 791 91 21 365 linda@nordlinkcapital.com EUROPE - NORDICS	● Michael Chambers 65 9858 9759 michael.chambers@aletheia-capital.com ASIA - SINGAPORE	● Graeme Bateman 852 2534 7437 graeme.bateman@aletheia-capital.com ASIA - HONG KONG
● Terrence Alford 1 469 403 9936 terrence.alford@aletheia-capital.com AMERICAS - DALLAS	● Daren Riley 44 748 31 59 865 daren.riley@aletheia-capital.com EUROPE - LONDON	● Richard Wallace 852 2534 7429 richard.wallace@aletheia-capital.com ASIA / EUROPE - HONG KONG	● Kevin O' Connor 61 431 095 979 kevin.oconnor@aletheia-capital.com AUSTRALIA - SYDNEY
● Dhananjay Mahurkar 44 777 55 26 870 dhananjay.mahurkar@aletheia-capital.com EUROPE - LONDON			

Firm Disclosures

Aletheia Capital Ltd ("Aletheia") is a limited company registered in Hong Kong, company number 2657920, located at Level 28 AIA, 1 Connaught Road, Central, Hong Kong.

Aletheia Analyst Network Ltd ("AAN") is a limited company registered in Hong Kong and is a wholly owned by Aletheia and is regulated by the Hong Kong Securities and Futures Commission, is a registered investment advisor with the U.S. Securities and Exchange Commission and is regulated by the Financial Conduct Authority, Firm Reference Number 794762.

Aletheia Capital (Singapore) Pte Ltd ("ACSG") is a limited company registered in Singapore, UEN 201823248E, and is a wholly owned by AAN and is an Exempt Financial Adviser as defined in the Financial Advisers Act and regulated by the Monetary Authority of Singapore.

This report was published by AAN and is distributed by AAN and ACSG. For investors in Singapore, this material is provided by ACSG pursuant to Regulation 32C of the Financial Advisers Regulations. If there are any matters arising from, or in connection with this material, please contact ACSG, 1 Fullerton Road, #02-01, One Fullerton, Singapore 049213, Tel: +65 6721 7116.

Additional information will be made available upon request.

Analyst Certification and Disclosures

The analyst(s) named in this report certifies that (i) all views expressed in this report accurately reflect the personal views of the analyst(s) with regard to any and all of the subject securities and companies mentioned in this report and (ii) no part of the compensation of the analyst(s) was, is, or will be, directly or indirectly, related to the specific recommendation or views expressed by that analyst herein.

The analyst(s) named in this report (or their associates) does not have a financial interest in the corporation(s) mentioned in this report.

Disclaimer

This report is for the use of intended recipients only and may not be reproduced, in whole or in part, or delivered or transmitted to any other person without our prior written consent. By accepting this report, the recipient agrees to be bound by the terms and limitations set out herein. The report is prepared for professional investors only whose business involves the acquisition, disposal or holding of securities, whether as principal or agent and must not be re-distributed to retail clients. The information contained in this report has been obtained from public sources believed to be reliable and the opinions contained herein are expressions of belief based on such information. Except as expressly set forth herein or in recipient's subscription agreement, no representation or warranty, express or implied, is made that such information or opinions is accurate, complete or verified and it should not be relied upon as such. Nothing in this report constitutes a representation that any investment strategy or recommendation contained herein is suitable or appropriate to a recipient's individual circumstances or otherwise constitutes a personal recommendation. It is published solely for information purposes, it does not constitute an advertisement, a prospectus or other offering document or an offer or a solicitation to buy or sell any securities or related financial instruments in any jurisdiction. Information and opinions contained in this report are published for the reference of the recipients and are not to be relied upon as authoritative or without the recipient's own independent verification or taken in substitution for the exercise of the recipient's own judgement. All opinions contained herein constitute the views of the analyst(s) named in this report, they are subject to change without notice and are not intended to provide the sole basis of any evaluation of the subject securities and companies mentioned in this report. Any reference to past performance should not be taken as an indication of future performance. Except in the case of fraud, gross negligence or willful misconduct, or except as otherwise set forth in recipient's subscription agreement, the firm does not accept any liability whatsoever for any direct or consequential loss arising from any use of the materials contained in this report.

© 2026 Aletheia Capital Ltd